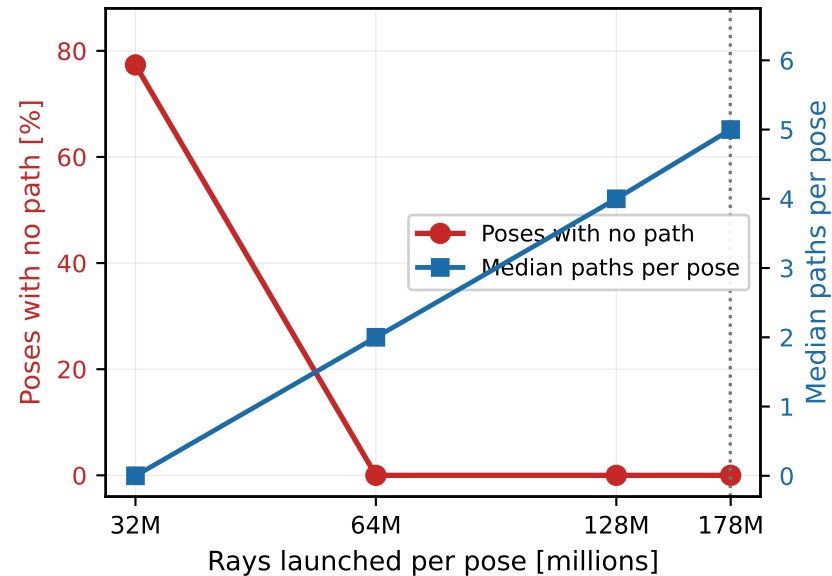
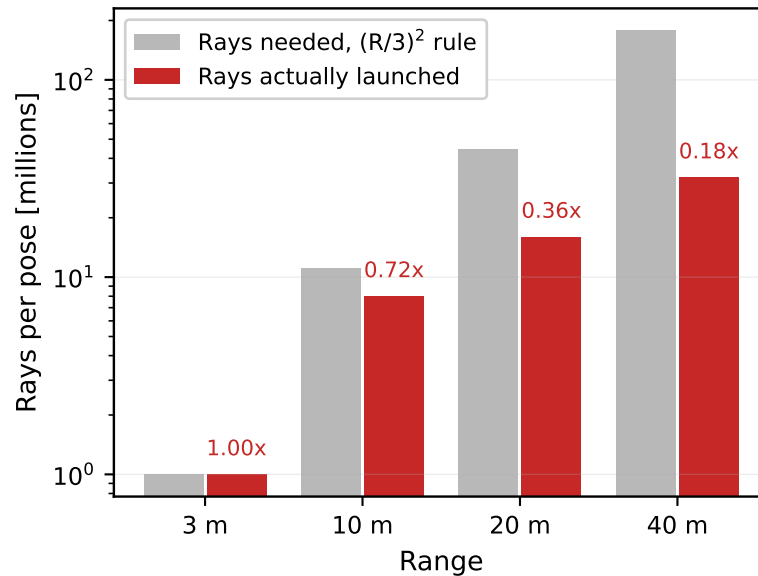


DJI Matrice 4E hovering, belly view, 3.5 GHz. Is range a wall for the path solver, or a budget?

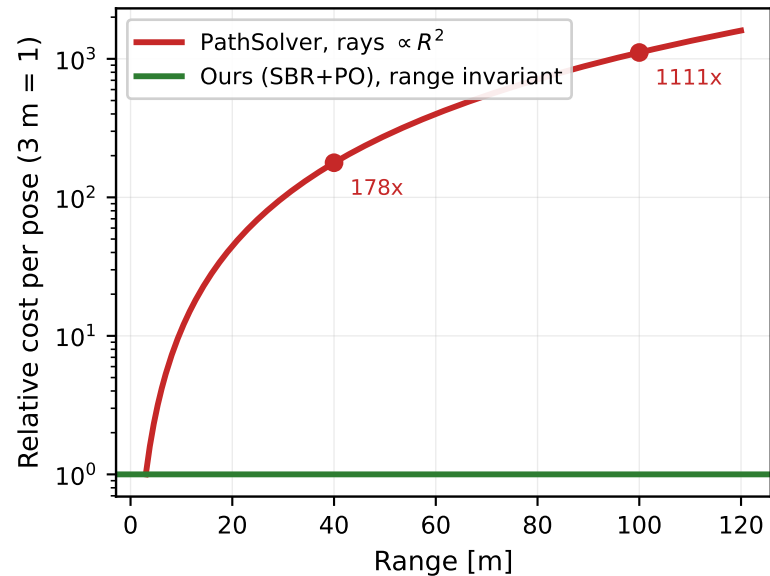
(a) At 40 m, rays alone fix it
77% empty at 32M, 0% at 64M



(b) The old sweep under-launched, and the
gap widens with range



(c) What actually remains is cost, and it
grows while ours does not



(a) Range, geometry, poses and rotor speeds are all held fixed at 40 m. Only the ray count changes, and the empty poses vanish, so the earlier collapse was a budget shortfall on our side rather than a property of the solver. (b) The dashed rule is rays scaled as $(R/3)$ squared from the 3 m case. (c) Cost is the honest limit. Holding the path count needs rays growing as R squared, and a micro-Doppler run stacks thousands of poses on top of that, while our target-anchored plane-wave grid is computed once and does not depend on range.